Fn of hardware

Hazards ≠

RAW, WAR, WAW

Fn of program

Dependencies

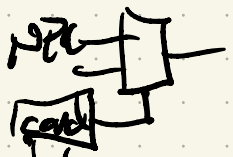
Pure/Flow, Anti-, output dependencies

```
ADD R1, R2, R3
SUB R4, R1, R5
```

+

Stall

Control Hazards



A: BEQ R1, G

B:

G

H

I

J

IF	ID	EX	MEM	WB
	IF	ID	EX	MEM
		IF	ID	EX
			IF	ID
				IF

X

MEM

Branches happen ~ 5 instructions

Don't do anything, best CPI = 1 + stalls

= 1.2

Bug: Control hazard

Feature it!

Machine/Assembly Language:

A branch doesn't take effect until 1 after the next instruction

Orig code

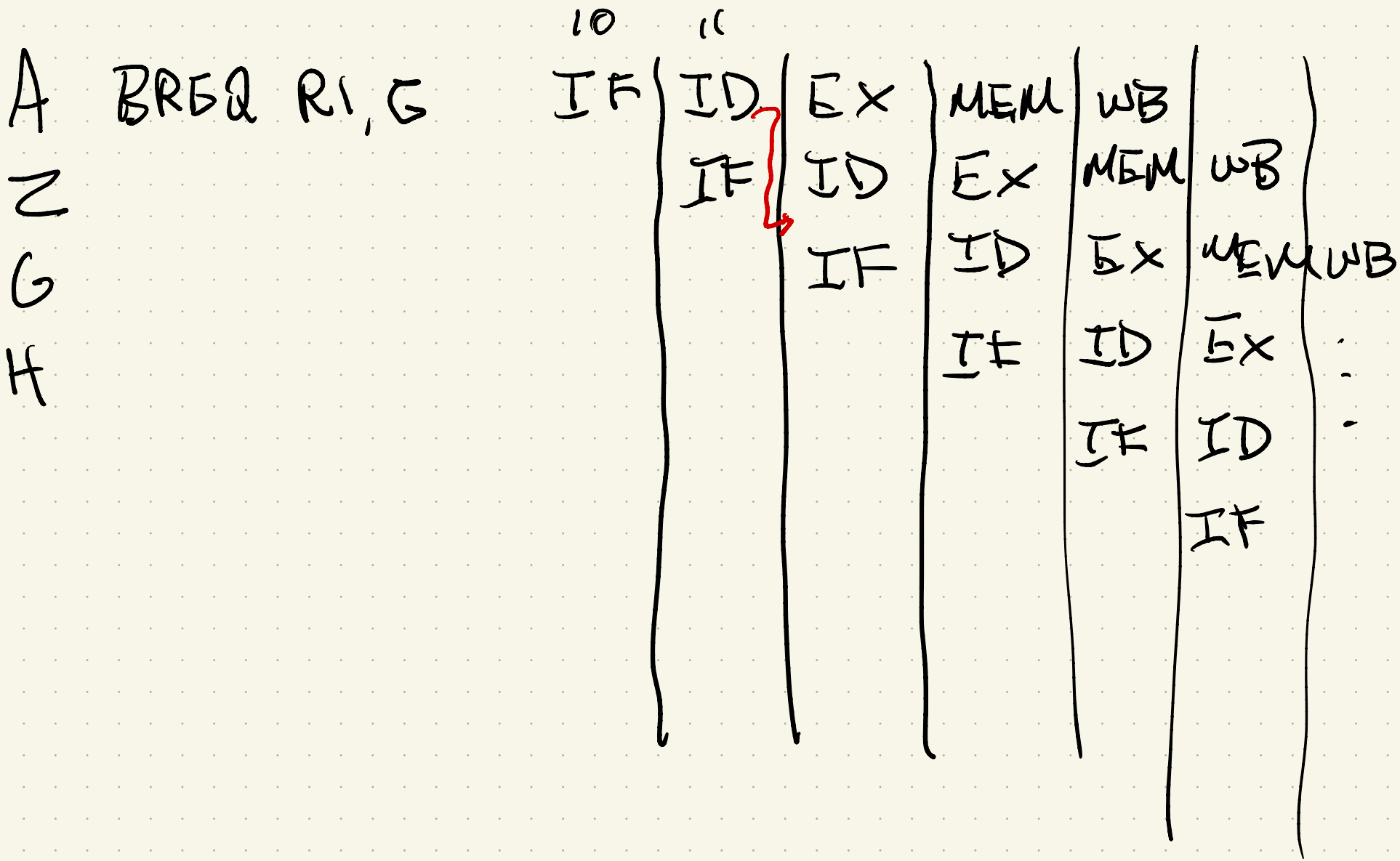
```
Z  ADD R4, R2, R3
A  BEQ R1, X
B  :
C  :
D  :
```

"Delay slot"

Corrected code:

```
A: BEQ R1, X
Z: ADD R4, R2, R3
B  :
C  :
D  :
```

Delayed branches!


10
11
A
BREQ R1, G
Z
G
H
IF
ID
EX
MEM
WB
IF
ID
EX
MEM
WB
IF
ID
EX
MEM
WB
IF
ID
EX
IF
ID
IF

Delay slots:

IF ID EX MEM WB

1 delay slot

IF1 IF2 ID1 ID2 EX1 EX2 MEM1 MEM2 WB

need next PC

know end of ID1

2 delay slots needed!

Before filling delay slots

```
ADD R1, R2, R3
BREQ R1, X
```

→

After

```
ADD R1, R2, R3
BREQ R1, X
NOP
```

1 slot in ISA 75-80% time
2 slots in ISA both ~ 30%
>2 slots ~ 0% of time

Fill slots from branch target

```
BREQ R1, G
   .
   .
   .
G: SUB R4, R5, R6
H:
I:
```

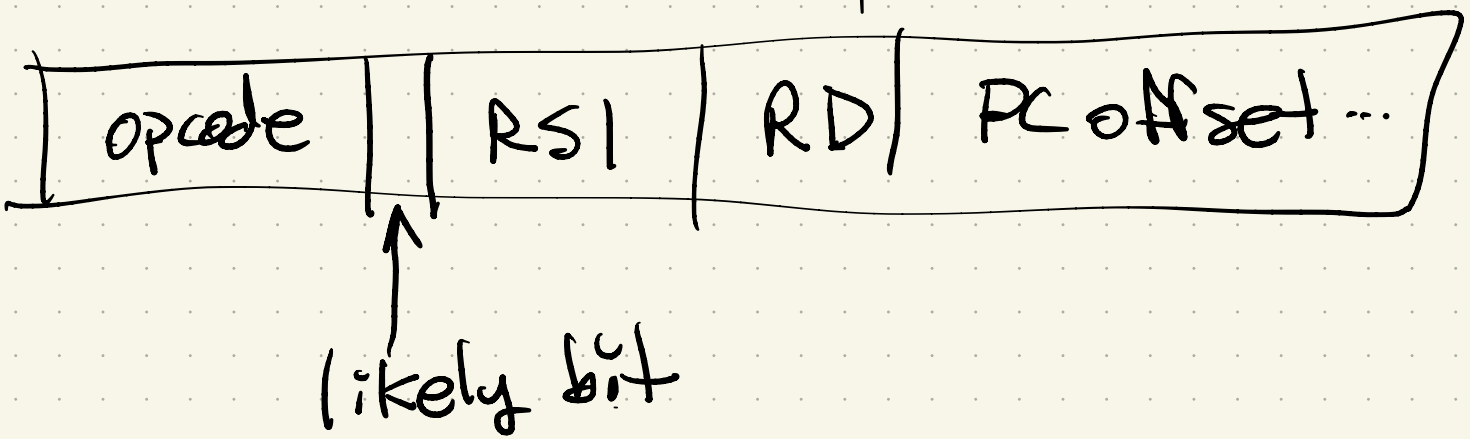
⇒

```
BREQ R1, H   (likely)
SUB R4, R5, R6
H:  .
I:  .
```

Squash slots

Two kinds of branches

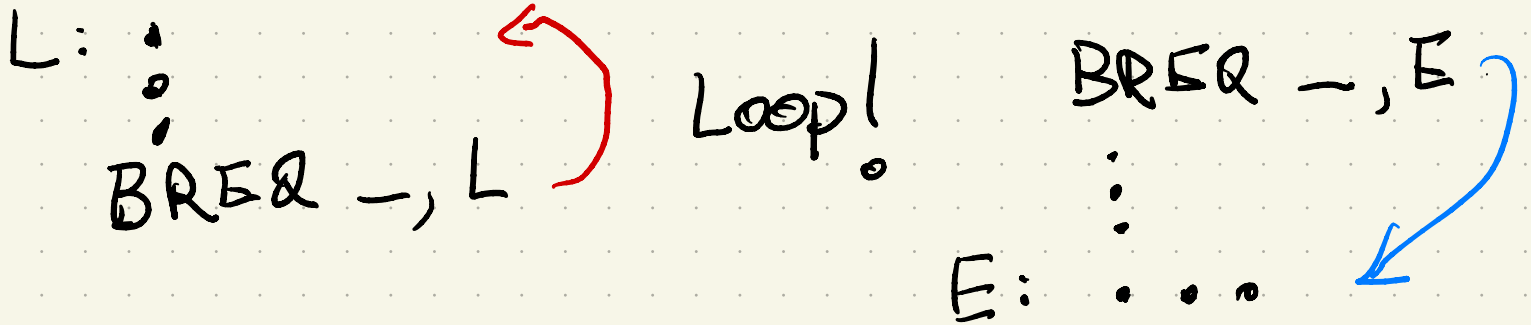
BREQ-likely R1, H | BREQ-unlikely R1, G



if likely bit	Squash when
= 1	branch not taken
= 0	branch taken

Methods to set likely bit

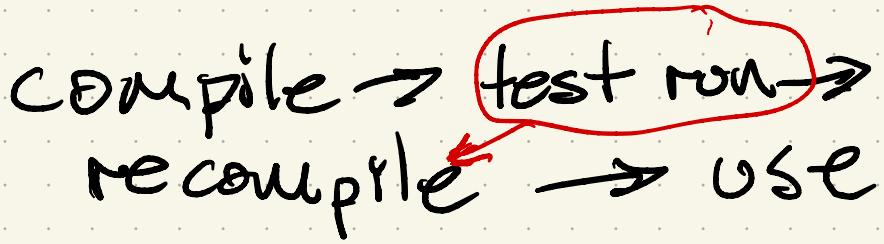
① Forward not taken, backward taken



② Heuristic

Ball/Larus

③ Profiling



Run time = IC × CPI × CT

IF1 IF2 ID1 ID2 EX1 EX2 MEM1 MEM2 WB

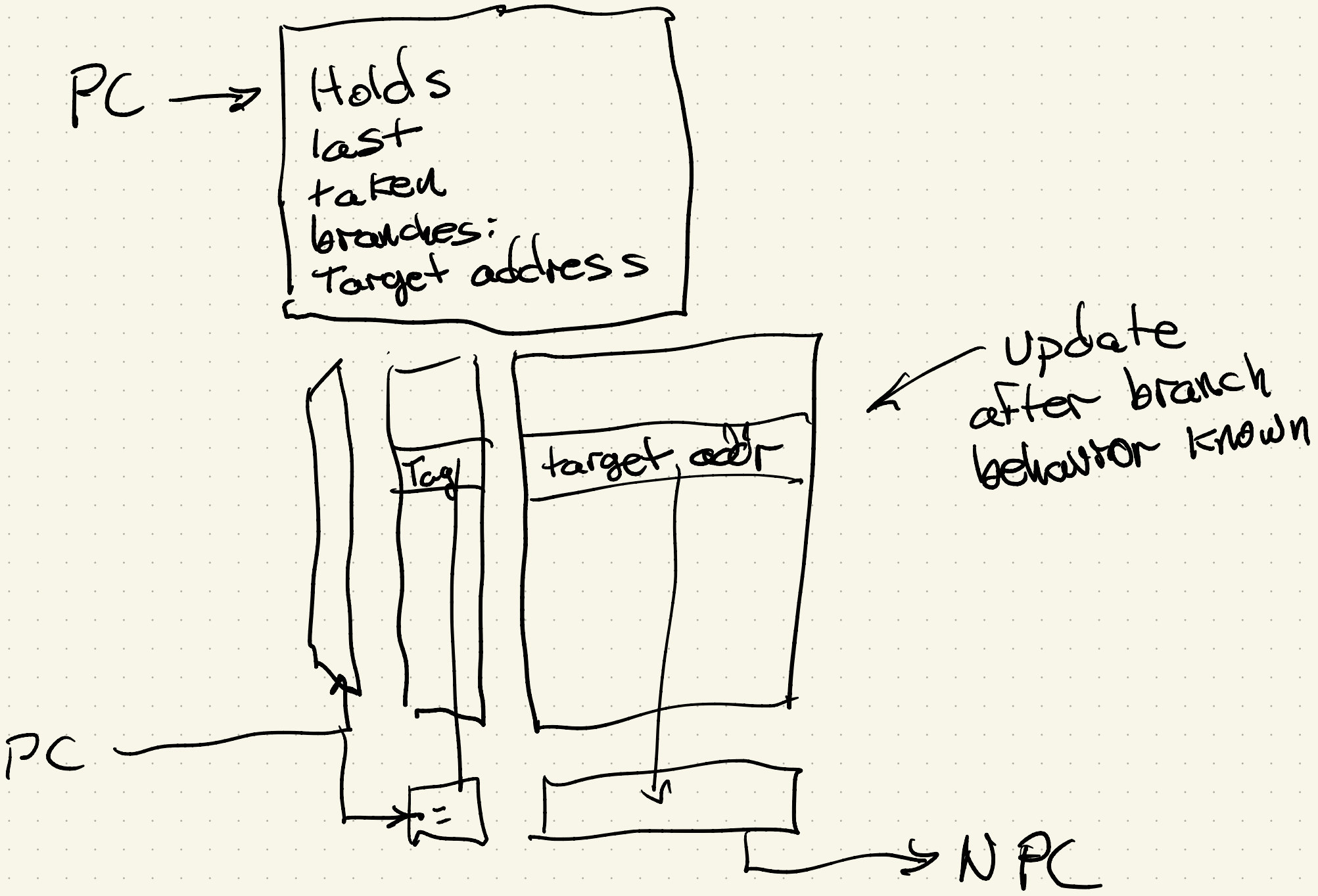
branch behavior need-to-know point

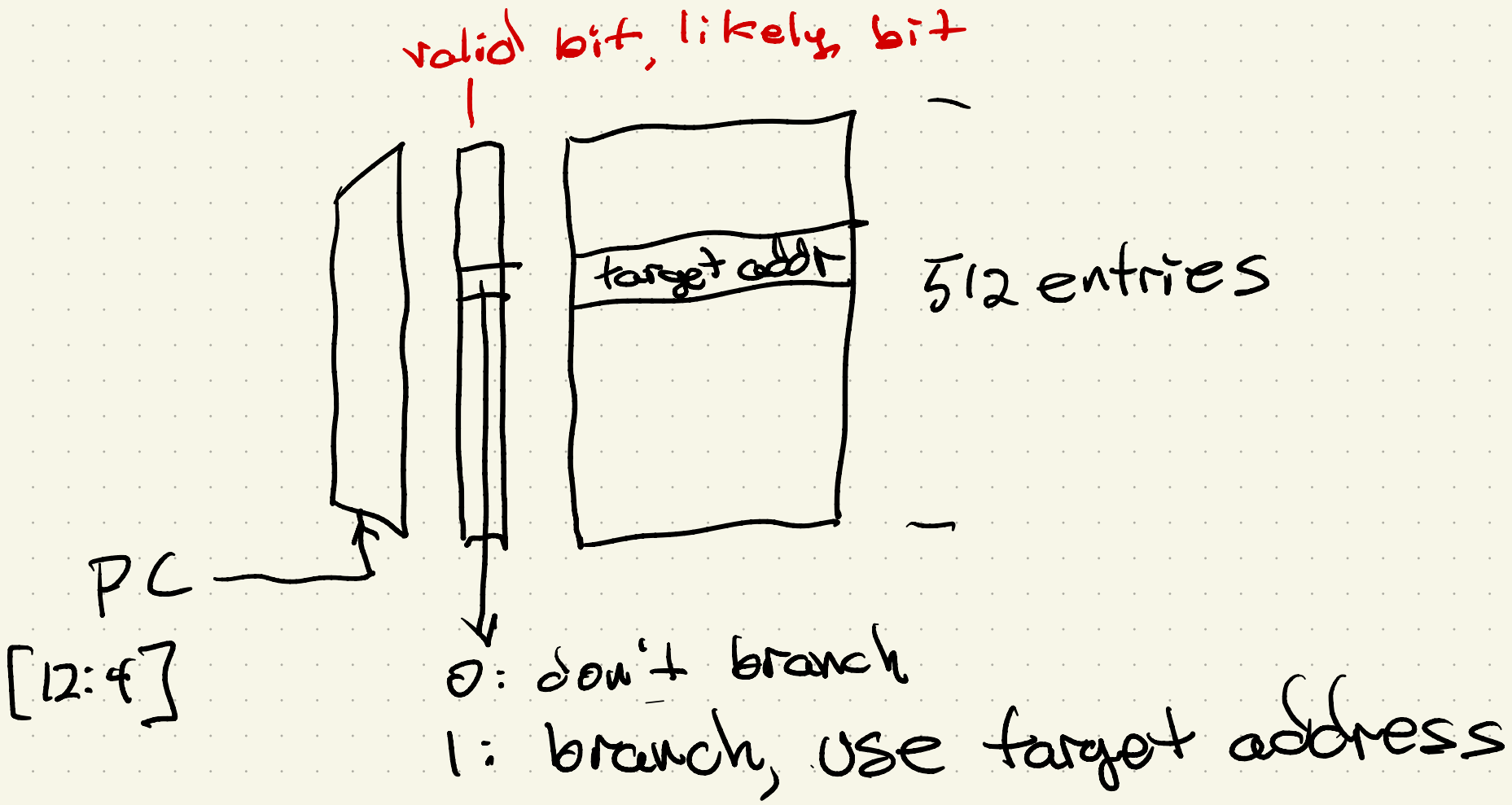
branch behavior known point

Need to know:

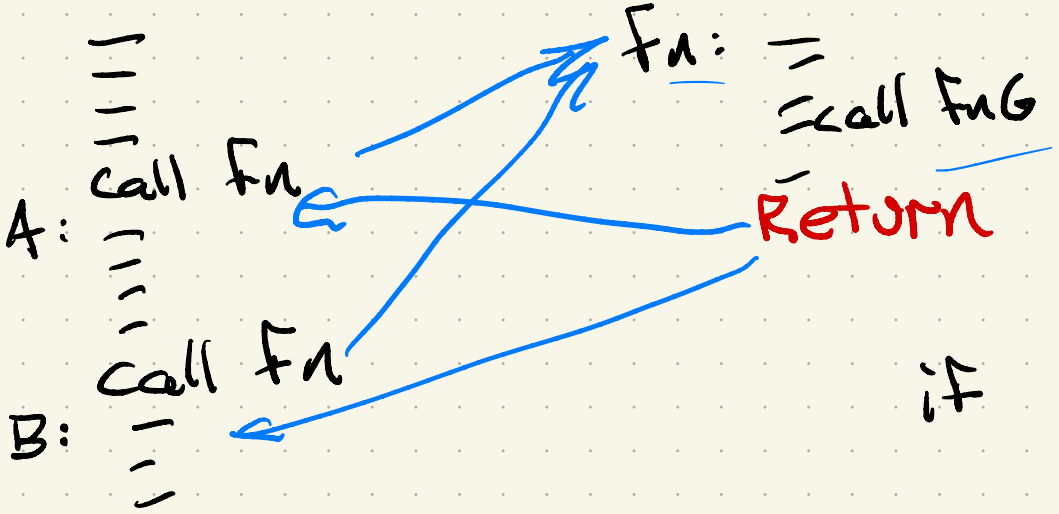
1. when instruction is a branch
2. where branch branches to
3. whether it branches or not

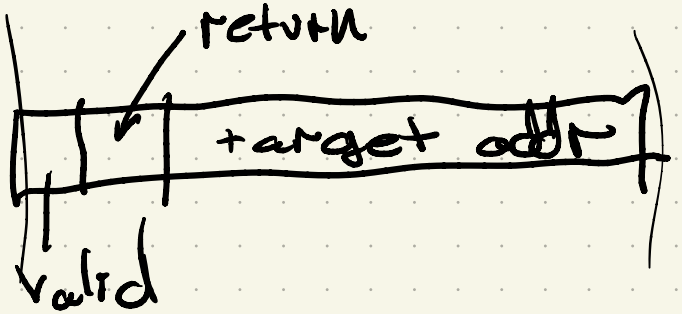
Branch Target buffer




valid bit, likely bit
target addr
512 entries
PC
[12:4]
0: don't branch
1: branch, use target address

Branches with unknown (indirect) Targets

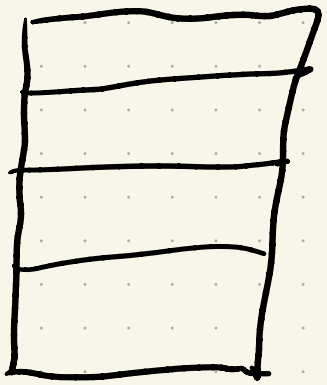


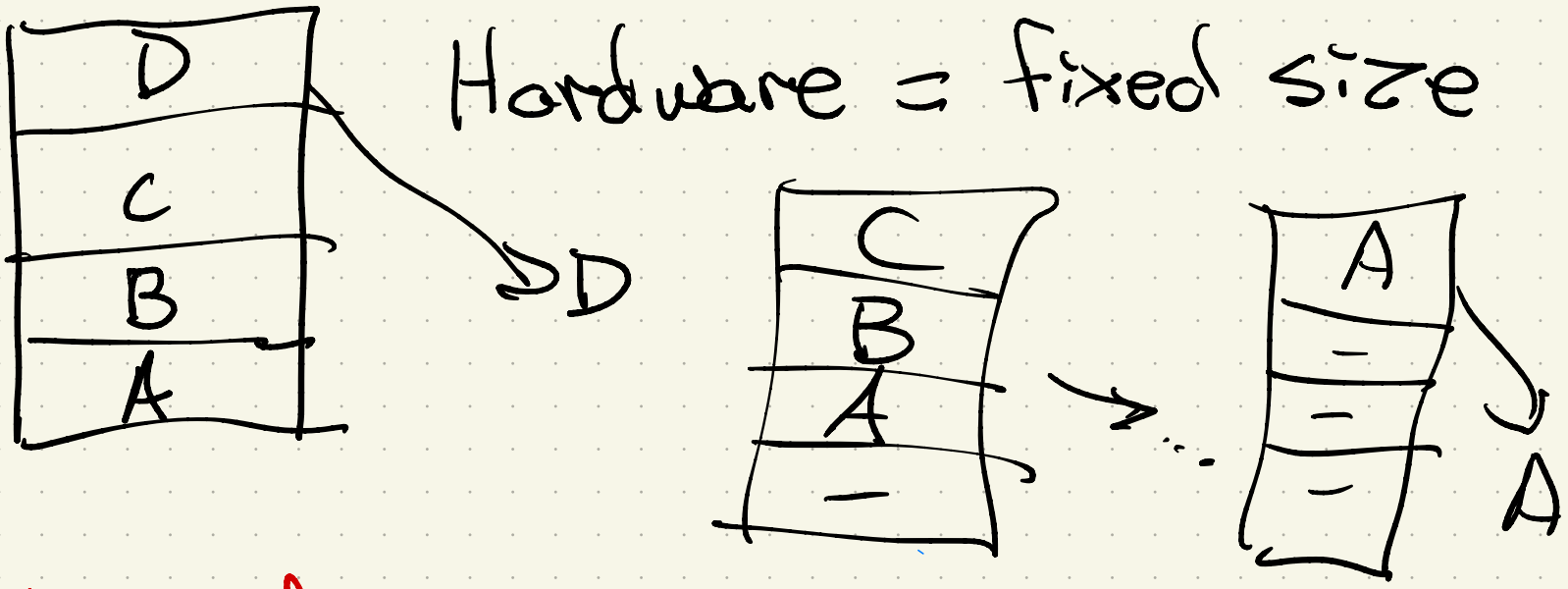


if return = (use hardware stack instead of branch target buffer

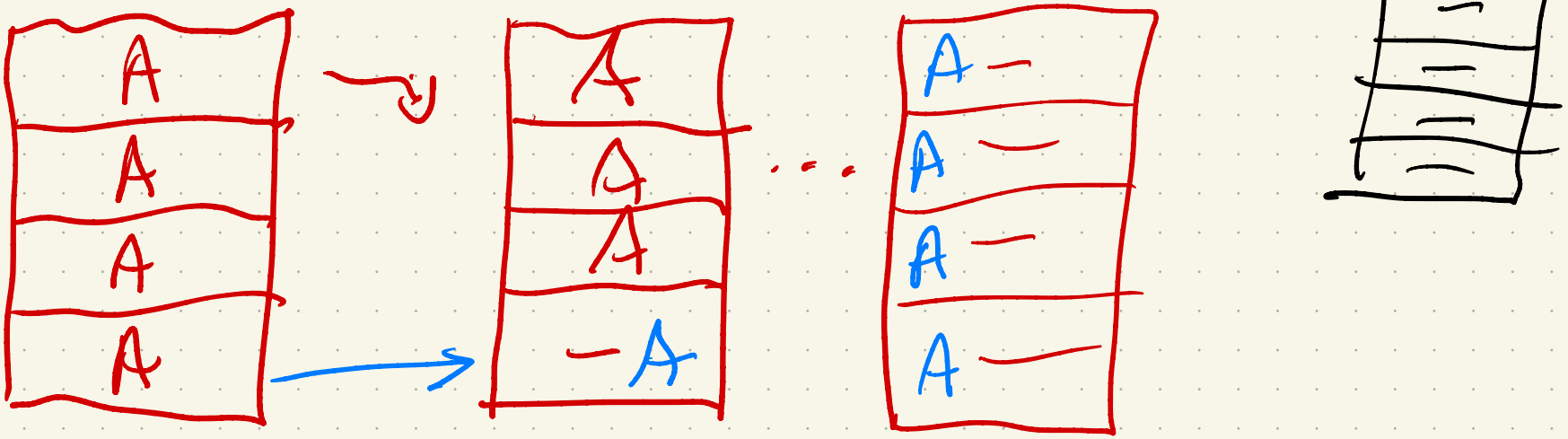
on call, push

on return, pop





only have deep stack is due to recursion



Duplicate bottom of stack on pop